

Wind-Climate Estimation Based on Mesoscale and Microscale Modeling: Statistical–Dynamical Downscaling for Wind Energy Applications

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ABSTRACT

This paper demonstrates that a statistical–dynamical method can be used to accurately estimate the wind climate at a wind farm site. In particular, postprocessing of mesoscale model output allows an efficient calculation of the local wind climate required for wind resource estimation at a wind turbine site. The method is divided into two parts: 1) preprocessing, in which the configurations for the mesoscale model simulations are determined, and 2) postprocessing, in which the data from the mesoscale simulations are prepared for wind energy application. Results from idealized mesoscale modeling experiments for a challenging wind farm site in northern Spain are presented to support the preprocessing method. Comparisons of modeling results with measurements from the same wind farm site are presented to support the postprocessing method. The crucial element in postprocessing is the bridging of mesoscale modeling data to microscale modeling input data, via a so-called generalization method. With this method, very high-resolution wind resource mapping can be achieved.

1. Introduction

A wind atlas is a geospatial dataset containing information about the wind climate that can be used to determine wind power resources at wind turbine sites. A wind atlas, typically covering 10^4 – 10^6 km², describes direction-sector-dependent wind speed distributions at a number of different heights above ground level (AGL) and a number of surface roughness types. A wind atlas can be derived from observations; however, where there is an insufficiently dense network of high quality wind measurements, this is not possible. Instead, methodologies based on numerical models have been developed to obtain geospatial wind climate data. One such methodology

is the Karlsruhe Atmospheric Mesoscale Model (KAMM) with the Wind Atlas Analysis and Application Program (WAsP) method in its earliest form first described in [Frank and Landberg \(1997\)](#) in which statistical–dynamical downscaling is used ([Frey-Buness et al. 1995](#)). This is the focus of this paper.

In the KAMM–WAsP method, the large-scale meteorological situations are derived from the National Centers for Environmental Prediction–National Center for Atmospheric Research (NCEP–NCAR) reanalysis dataset ([Kalnay et al. 1996](#)) but could be from any other reanalysis dataset. The NCEP–NCAR dataset is used to create between 100 and 150 different large-scale wind situations, called wind classes, that represent the large-scale wind climate. KAMM ([Adrian and Fiedler 1991](#)) determines how the large-scale wind forcing of each wind class is modified by regional-scale topography. In the KAMM–WAsP method, postprocessing of the results from all the simulations yields a wind resource map at the resolution of the model simulations. Further

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analysis, with consideration given to the mesoscale model description of topography, yields wind atlas maps for generalized surface conditions (flat terrain with uniform roughness). This generalized wind climate is a unique feature of the methodology. The generalized wind climate approach allows the wind climate to be downscaled further via microscale models.

Statistical–dynamical downscaling has a number of advantages over dynamical downscaling. Usually, a statistical–dynamical downscaling approach is more economical in terms of computer resources. This can liberate computer resources for mesoscale simulations at a higher resolution or permit a greater degree of sensitivity testing, compared to the use of dynamical downscaling alone. Also, the approach can permit the recalculation of wind statistics for shorter, specific time periods without requiring new mesoscale simulations, if the simulations adequately cover the full climatological distribution of the wind conditions.

Yu et al. (2006) presented a wind-mapping tool method based on the KAMM–WASP methodology. However, there are important differences between Yu et al. (2006) and the method described in this article on how the geostrophic wind classes are defined and significant differences on how postprocessing of the mesoscale model output is performed. Although this article and Yu et al. (2006) use different mesoscale and microscale models, the main aspect to consider is the framework within which these models are used and linked in order to perform downscaling.

Bergström (2001) presented a statistical–dynamical downscaling methodology using the Meteorological Institute of the University of Uppsala model (MIUU model) run employing 96 forcings (three speeds, eight directions, and four seasons). In that methodology there is no linkage to microscale modeling; thus, the results can only be interpreted at the 9-km resolution of the MIUU model mesoscale simulation. Also, no explicit verification of the model is given, except for some broad comparisons of modeled and observed winds. Mengelkamp et al. (1997) presented a statistical–dynamical downscaling methodology using 143 forcings based on a cluster analysis of radiosonde data. Winds at 850 hPa and temperature difference between 100 and 1500 m were used as the basis for the cluster analysis. No linkage to microscale modeling was carried out. Verification is performed using 10-m wind data. The conclusions of that paper suggested the importance of flow influences at the point of measurement, which reinforces the motivation for a linkage between the mesoscale and microscale modeling approaches. Pinto et al. (2010) also presented a statistical–dynamical downscaling, this time for assessing storm impacts in a changing climate. Core to the

methodology is an assessment of synoptic development conditions by the K-means cluster algorithm, giving 55 different clusters (Leckebusch et al. 2006). Such a clustering of synoptic situations could also be applied to numerical wind atlas calculations but, as will be discussed later, the wind class method in this paper has some advantages over a more synoptic situation weather typing approach.

The outline of the paper is as follows. In section 2, a description is given on how wind classes are generated. Next, in section 3, a description of the KAMM mesoscale model is given, followed by its application in idealized modeling experiments, which support the wind class methodology. The generalization method applied to mesoscale model winds is explained in section 4. The application of the generalization procedure and validation against measurements are given in section 5. In section 6, a discussion about uncertainty and other applications is given. Conclusions are drawn in section 7.

2. Preprocessing: Wind class generation

To introduce the method, an example applying the entire methodology is given for the Alaiz wind farm site in the Navarra region of northern Spain. The site location has been studied extensively by several modeling groups, starting with Giebel et al. (2006). The location of the Alaiz site is shown in Fig. 1.

NCEP–NCAR reanalysis data on isobaric surfaces, on a latitude–longitude grid with grid spacing of $2.5^\circ \times 2.5^\circ$, were used to determine the large-scale meteorological situation in the region of the site. At 43.75°N , 1.75°W the geostrophic wind, virtual potential temperature, and specific humidity were calculated at heights of 0, 1500, 3000, and 5500 m above mean sea level using a linear interpolation procedure on data on 1000-, 850-, 700-, and 500-hPa isobaric surfaces at the surrounding reanalysis points: 42.5°N , 2.5°W ; 42.5°N , 0.0° ; 45.0°N , 2.5°W ; and 45.0°N , 0.0° . These points are shown in Fig. 1. The geostrophic wind speed for 2001 at 0 m as a function of direction is shown in Fig. 2. That year is chosen because measurement data were available at the site. However, for climatological studies the practice is to use 30 yr of NCEP–NCAR reanalysis data. The issue of time period is returned to in section 6a.

Using reanalysis data for the forcing of mesoscale models is a well-established technique (Pinto et al. 2010; Yu et al. 2006), whereas Mengelkamp et al. (1997) used radiosonde data for the basis of the mesoscale simulations. The main issue with radiosonde wind data is that it could be affected by the terrain in the vicinity of the radiosonde station. Furthermore, the relevant local terrain

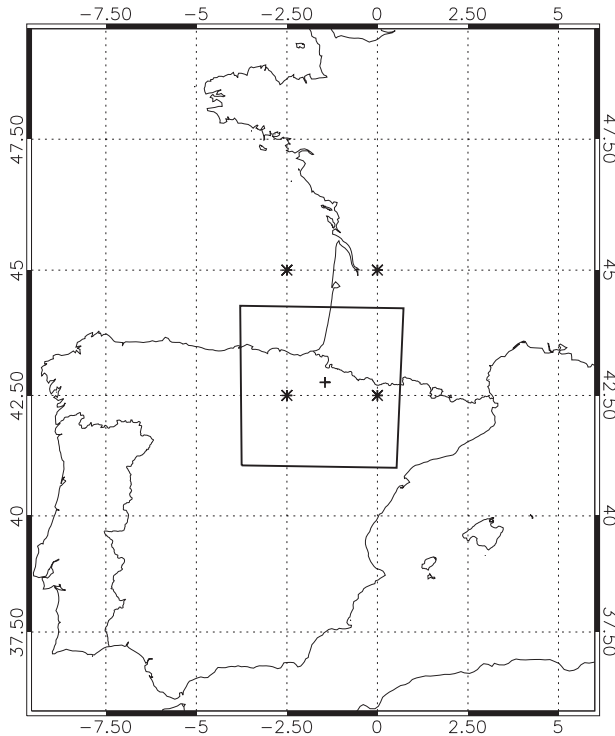


FIG. 1. The location of the Alaiz site (plus sign) within the larger context of western Europe. The asterisks mark the NCEP-NCAR grid points used to determine the wind classes. The inner box indicates the mesoscale modeling domain.

impacting the radiosonde ascent will depend on the exact trajectory of the radiosonde.

The process of generating wind classes divides the reanalysis profile data into bins, defined by wind direction, wind speed, and Froude number. The Froude number Fr is

$$Fr = \frac{U}{hN}, \quad (1)$$

where U is the wind speed, h is a height scale, and N is the Brunt–Väisälä frequency,

$$N = \sqrt{\frac{g}{\theta_{v_0}} \frac{\partial \theta_v}{\partial z}}, \quad (2)$$

where θ_{v_0} is the mean virtual potential temperature, $\partial \theta_v / \partial z$ is the vertical gradient of virtual potential temperature, and g is the gravitational acceleration. The Froude number (shown in colors in Fig. 2) is used in the wind class classification system to differentiate thermal stratification. For cases in which Fr is below 1, winds tend to flow around obstacles. For cases in which Fr is above 1, winds tend to flow over obstacles or orography. From (1), more stable conditions or lower wind speed situations

give lower Froude number flow, and thus channeling of winds between or around obstacles in the terrain (i.e., hills and mountains) is more prevalent. For less stable conditions or higher wind speed situations, giving a higher Froude number, the same terrain features present less of an obstacle to the flow.

Wind classes were defined by first binning the profile data into 12 equal-sized direction sectors according to the geostrophic wind at sea level. Then, for each direction sector, the data were divided into five wind speed bins. Some of these wind speed bins were further split into two bins according to Froude number, the bisecting Fr value being the median value of the Froude number for the wind speed bin in question. A maximum of 10 wind class bins (defined by wind speed and Froude number) were accommodated per direction sector. The actual number of wind classes per direction sector was determined by setting a minimum frequency allowed for a wind class (0.44%). The wind speed bin limits were defined so that wind class frequencies within a direction sector were equal, except for the lowest and highest wind class bins, which had lower frequencies in order to represent better the tails of the wind speed distribution. In this study, for a given direction sector the upper limit of the lowest wind speed bin is the 19th percentile value and the lower limit of the highest wind speed bin is the 91st percentile value of the wind speed distribution in that direction sector. Away from the tails of the distribution, the wind speed bins have widths corresponding to a 23.8% share of the wind speed distribution in that direction sector.

This near-equal weighting, across wind classes, means that the results from each numerical simulation have near-equal weight, and hence the computational effort is distributed evenly over the different meteorological situations. The goal of the wind class method is to capture the large-scale direction-sector-dependent wind speed distributions to an appropriate level of accuracy. As an accuracy check, the wind climate distribution statistics derived from wind classes, and their associated weightings, are compared with statistics calculated directly from the NCEP–NCAR time series data over the same period. In this case with 108 wind classes, the mean geostrophic wind speed is determined by the wind classes to within 0.01% of the value calculated directly from the NCEP–NCAR time series, and the mean of the geostrophic wind speed cubed is determined to within 7%, at the level where the classes are defined. When the maximum number of wind speed bins per direction sector is increased from 5 to 10, the mean of the geostrophic wind speed cubed is determined to within 3%. On the other hand, when the number of wind speed bins per direction sector is decreased from 5 to 3, the

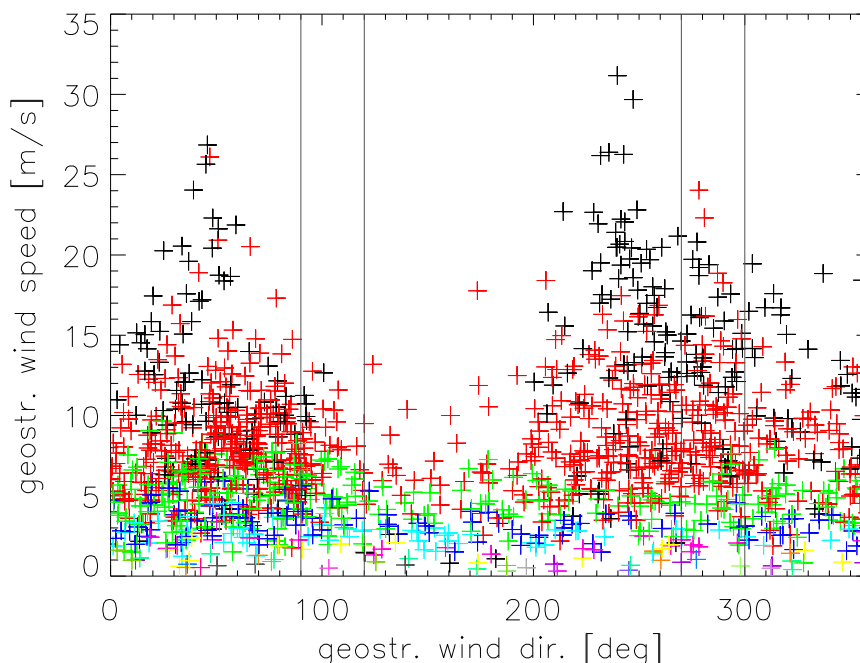


FIG. 2. Geostrophic wind speed vs geostrophic wind direction for 2001 derived from NCEP–NCAR reanalysis data surrounding the Alaiz site. The colors indicate Fr: $1 \leq \text{Fr}$, black; $0.58 \leq \text{Fr} < 1$, red; $0.45 \leq \text{Fr} < 0.58$, green; $0.38 \leq \text{Fr} < 0.45$, blue; $0.33 \leq \text{Fr} < 0.38$, light blue; $0.3 \leq \text{Fr} < 0.33$, purple; and $0.28 \leq \text{Fr} < 0.33$, yellow. The vertical lines indicate the direction ranges referred to in the text.

mean of geostrophic wind speed cubed is determined within 13%.

Figure 3 shows a representation of the wind classes, pertaining to the site. As in Fig. 2, it also shows the Froude number using a color scale. Altogether, 108 wind classes were generated. The height scale used, h in Eq. (1), is 1500 m, the height difference between the first and second level in the wind class profile. This height scale is approximately equal to the height of the nearby terrain. If the modeling were performed over a flatter domain, the height scale might be reduced. However, a height scale of 1500 m was also used for a modeling study over Denmark that produced good agreement with measurements (Frank et al. 2001). The virtual potential temperature gradient was calculated between 0 (sea level) and 1500 m, and the geostrophic wind velocity was calculated at 0 m (sea level). The Froude number based on geostrophic wind is advantageous over the real wind speed from the reanalysis because it is not sensitive to the roughness length description in the reanalysis model and it is consistent within the framework of geostrophic wind forcing used for the mesoscale modeling. In the Alaiz example, the most frequent geostrophic winds at sea level were from the northeasterly and westerly sectors, and it is from these direction sectors that the geostrophic winds were the strongest. In this

wind class set, the minimum allowed frequency for a wind class was 0.44%. This value is adjusting as a way to limit the total number of wind classes. It is especially useful as a way to limit the number of wind classes in direction

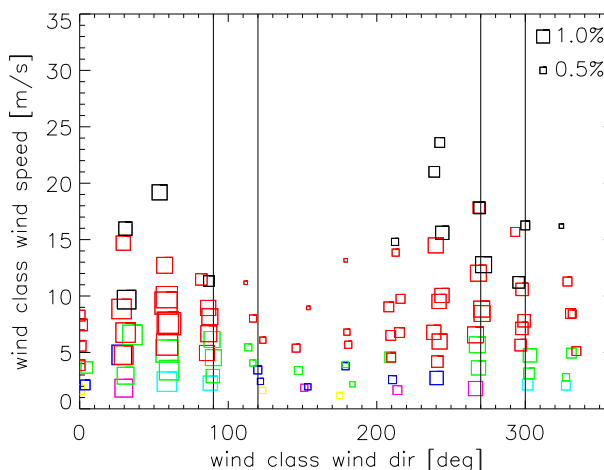


FIG. 3. Wind classes for the Alaiz mesoscale modeling study based on the data shown in Fig. 2. The size of the square represents the frequency of occurrence of the wind class; largest with frequency 1.712%, and smallest with frequency 0.274%. The colors indicate the Fr, as in Fig. 2, and the vertical lines indicate direction ranges referred to in the text.

sectors with a low frequency of occurrence. For example, in the direction sector in the range $105^\circ < \theta < 135^\circ$ with a frequency of 2.53%, only five wind classes were defined. For a high-frequency direction sector, such as in range $255^\circ < \theta < 285^\circ$, the maximum of 10 wind speed classes were defined.

3. Mesoscale modeling

a. The KAMM model

KAMM is a nonhydrostatic mesoscale model and is described in [Adrian and Fiedler \(1991\)](#). Spatial derivatives are calculated in the model by central differences on a terrain-following grid. The turbulent fluxes are modeled using a mixing-length model with stability-dependent turbulent diffusion coefficients in stably stratified flow, and a nonlocal closure for the convective mixed layer. Lateral boundary conditions assume zero gradients normal to the inflow sides. On outflow boundaries, the horizontal equations of motion are replaced by a simple wave equation allowing signals to propagate out of the domain without reflection. Gravity waves can penetrate the upper boundary outward using the boundary conditions of [Klemp and Durran \(1983\)](#).

KAMM is able to run as a “stand alone” model; that is, the model can be initialized using only a large-scale forcing in the form of a single vertical profile of geostrophic wind and virtual potential temperature. Thus, this profile of geostrophic wind applies over the whole domain. Hence, within this idealized framework, it is not necessary to nest the mesoscale model within a larger model that supplies boundary conditions. No radiative processes are modeled. The surface temperature is set equal to the lowest model air temperature at the initial time and remains constant throughout the integration.

In the vertical, the model extends from sea level to 5500 m above sea level, using 25 model levels. KAMM uses a terrain-following coordinate system in which the interval between vertical levels is not uniform, allowing for more closely spaced vertical model levels close to the terrain. The first five model levels are at 0, 20.3, 58.7, 115.3, and 190.0 m above the surface. The separation between these levels is smaller in elevated terrain. A horizontal grid spacing of 5 km is used. The choice of modeling domain requires the consideration of both the region of interest that must be covered by the domain and the regions outside the region of interest, which may impact the flow behavior within the region of interest.

b. Model setup and simulation results

The site has been the subject of numerous investigations, some of which have been conducted as part of the European Union (EU) funded “ANEMOS” project

([Kariniotakis et al. 2004](#)), because of challenges in providing accurate wind and power forecasts. The site is located on an approximately 1000-m-high mountain, located between regions of even higher elevated terrain to the east and west. [Figure 4](#) shows the elevation map of the calculation domain derived from the Global 30 arc s elevation dataset (GTOPO30; <https://lta.cr.usgs.gov/GTOPO30>). [Figure 5](#) shows the surface roughness for the calculation domain derived from Global Land Cover Classification (<http://edc2.usgs.gov/glcc/glcc.php>). The size of the domain is 400 km $\times$ 400 km in order to include the important topographical features influencing the flow in the region of the Alaiz wind farm.

The profiles of geostrophic wind speed and direction and virtual potential temperature for each of the wind classes described in [section 2](#) are used as forcing conditions in KAMM simulations, configured as described above. The modeled wind speed at 50 m AGL for the wind farm site is shown in [Fig. 6](#). For comparison, [Fig. 7](#) shows the observed distribution of wind speed and direction at 55 m AGL. The normalized values are given because the absolute values are sensitive and protected data. The direction-concentrating effect of the topography at the wind farm site is evident in both the KAMM simulations and the observations. The directions with highest winds and frequency (north-northwesterly and south-southeasterly) are reproduced. However, the strongest winds are from the northerly direction for the observations, whereas for the modeling results there is little difference between the northerly and southerly wind strengths. This point will be returned to and explained in [section 4](#). The rarity of wind direction measurements in the range $30^\circ \leq \theta \leq 120^\circ$ is mirrored by the near absence of modeled winds in that direction sector. The absence of high wind speeds in the direction range $210^\circ \leq \theta \leq 300^\circ$ is also seen in the modeling results.

c. Idealized mesoscale modeling

The observed and indeed modeled characteristics of the wind farm site, seen in [Figs. 7](#) and [6](#) respectively, show how drastically altered the surface winds are compared to the geostrophic winds. An idealized modeling approach is used to investigate further the flow behavior at the site. With an idealized approach, it will be easier to assess the role of geostrophic wind direction and speed, as well as the profile stability, in determining the surface winds at the wind farm site.

Simulations using KAMM have been performed using sets of idealized wind profiles representing different wind directions and different atmospheric stabilities. Four different geostrophic wind speed values (5, 10, 15, and 20 m s^{-1}) and three different stabilities (typical,

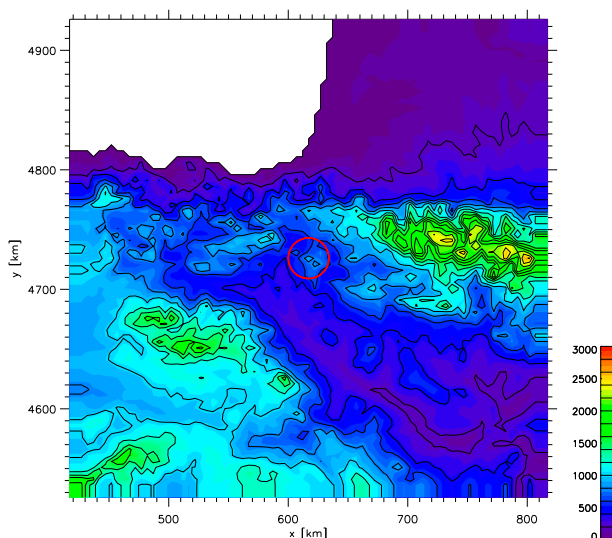


FIG. 4. Surface elevation (m) at 5-km resolution for the calculation domain for the Alaiz simulations. The axes are in UTM coordinates in kilometers [zone 30 datum from the 1984 revision of the World Geodetic System (WGS84)]. The contour interval is 200 m. The center of the red circle indicates the location of the wind farm site.

stable, and almost neutral) were investigated. The geostrophic wind and temperature profiles were defined at 0, 1500, 3000, and 5500 m above sea level. The geostrophic wind speeds and directions forcing the model were constant with height for each profile. For the stable and typical stability cases the temperature profiles were evaluated using NCEP–NCAR reanalysis data. Table 1 gives the virtual potential temperature profiles.

Figure 8 summarizes the results for five sets of the idealized KAMM simulations. The thick rectangles, around each box, show the forcing directions; for example, the direction sector centered on 30° is red. For a given large-scale forcing the simulated winds at 50 m AGL at the wind farm are shown by lines of the same color. The boldface line in Fig. 8 is at the farm site grid point and four thinner lines indicate the simulated wind locations neighboring the site: one grid point to the north, east, south, and west. The direction on the diagram indicates the direction where the wind comes from and the length indicates the wind speed normalized by the geostrophic wind forcing. The dotted-line circle represents the normalized wind speed value of 1, meaning that at this radius the wind speed at 50 m AGL is the same as the geostrophic forcing. A normalized wind speed value above 1 indicates a wind faster than the geostrophic forcing wind speed and a value below 1 indicates a wind slower than the geostrophic forcing wind speed.

The separate plots in Fig. 8 have been arranged so that when moving from left to right in the figure the

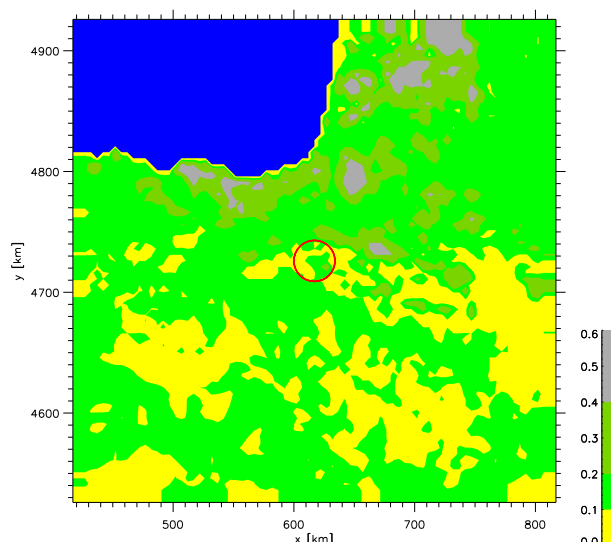


FIG. 5. As in Fig. 4, but for surface roughness length (m).

simulation sets have increasing Froude number, associated with the decrease in statistic stability. Also, when moving from the bottom to the top in the figure, the Froude number increases and is associated with the increased geostrophic wind. The large-scale Froude number characterizes the behavior of the modeled winds at the site. For 10 m s^{-1} geostrophic winds and typical stability (Fig. 8c), the wind directions at the wind farm site tend to be concentrated into the northern and southeastern sectors, whatever the forcing wind direction. For the set using 5 m s^{-1} geostrophic winds and typical stability (Fig. 8e), the direction concentration effect is slightly increased and there is a greater degree of enhanced wind speed at the site. A similar strengthened direction concentration and pattern of enhanced wind speed behavior is seen in the set using 10 m s^{-1} winds and stable stratification (Fig. 8b). For the set of simulations using 20 m s^{-1} winds and typical stratification (Fig. 8a), there is a reduction in the direction concentrating effect of the winds at the wind farm, and a reduced enhanced wind speed effect can be seen. For the set using 10 m s^{-1} winds and neutral stratification (Fig. 8d), there is a further reduction in the direction concentrating effect (cf. Fig. 8a) although the enhancement of the wind speed is similar.

To summarize: at the wind farm site, a lower Froude number flow tends to give more concentrated wind directions and a higher degree of enhanced wind speed compared to higher Froude number flows. In terms of the mesoscale wind dynamics and its interaction with the terrain, for lower Froude numbers the flow is forced to pass through the gap between the elevated terrain to the west and east, giving a north-northwesterly or

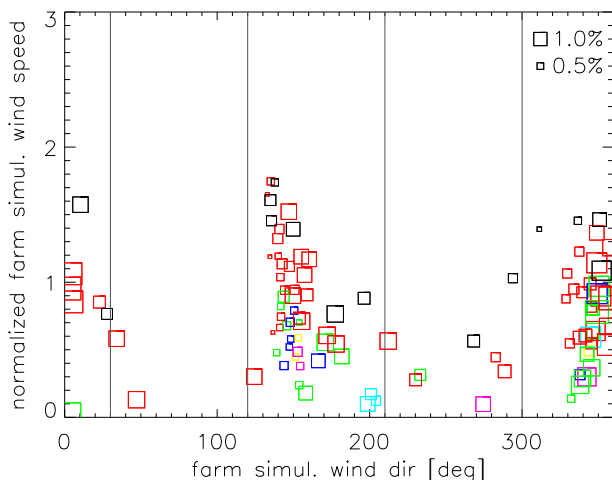


FIG. 6. Normalized simulated wind speed vs simulated wind direction at the Alaiz site for 2001. The normalization is obtained by dividing by the mean measured wind speed. The size of the square represents the frequency of occurrence of the wind class; largest with frequency 1.712%, and smallest with frequency 0.274%. The colors indicate the Fr, as in Fig. 2, and vertical lines indicate direction ranges referred to in the text.

south-southeasterly wind direction. For higher Froude number the flow is impacted less by the topography, and the elevated terrain is surmounted to a greater degree. There is still a signature of gap flow, particularly in the 20 m s^{-1} winds and typical stratification cases (Fig. 8a), but less so in the cases using 10 m s^{-1} winds and neutral stratification (Fig. 8d). The break angles are 105° and 285° . Similar to the low Froude number cases, the maximum wind speed enhancement is seen at $333^\circ \leq \theta \leq 60^\circ$ (cyclically) and $150^\circ \leq \theta \leq 210^\circ$. The geostrophic winds that are impacted most are $60^\circ \leq \theta \leq 90^\circ$ and $240^\circ \leq \theta \leq 270^\circ$, as these cases exhibit both a large turn in surface wind direction and large wind speed enhancement.

Figures 9a,b show the 50 m AGL winds for the modeling domain for geostrophic wind forcing of 10 m s^{-1} from 60° for stable and near-neutral conditions, respectively. For the stable conditions (Fig. 9a), it can be seen from the wind vectors that the flow direction is altered from northeasterly in the upper part of the domain to northwesterly in the central and southeastern parts of the domain. This diverted flow is associated with the Sierra de Cabollera and western shoulder of the Pyrenees. For the near-neutral conditions (Fig. 9b) the diversion of the wind direction associated with the terrain is much reduced. In contrast the wind direction over the entire domain is comparatively uniform compared to the stable conditions (Fig. 9a). In addition, the wind speed in the stable conditions is enhanced in the region of the gap flow, whereas this enhanced wind feature is absent in the near-neutral conditions. Similar behavior is seen,

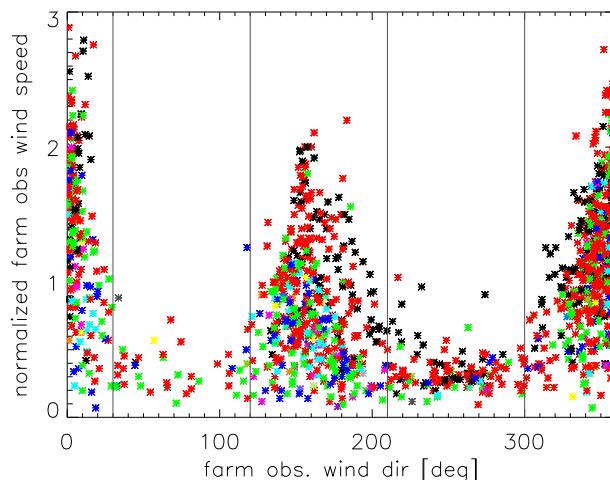


FIG. 7. As in Fig. 6, but for measured wind.

although with reversed wind direction, for the geostrophic wind forcing of 10 m s^{-1} at 240° for stable and near-neutral conditions, shown in Figs. 9c and 9d, respectively. For stable conditions, the flow is enhanced in the gap producing strengthened southeasterly winds, as compared with the much weaker channel flow in near-neutral conditions.

The wind farm site is placed on the northern end of the gap of elevated terrain, so when the gap flow has a southerly component, the wind farm site is at the exit region of the gap flow. Conversely, for a northerly component of the gap flow, the site is at the entrance. This explains the asymmetry of the two predominant wind directions; that is, the predominant wind directions are not 180° apart (twofold rotationally symmetric).

Geostrophic winds forcings with direction in the range $120^\circ \leq \theta \leq 270^\circ$ give south and southeasterly winds at the wind farm site. Conversely, geostrophic wind forcings with direction in the range $300^\circ \leq \theta \leq 90^\circ$ give northerly and northwesterly winds at the wind farm site. The enhanced wind speed effect is greatest for forcings with geostrophic wind direction $150^\circ \leq \theta \leq 210^\circ$ and $330^\circ \leq \theta \leq 60^\circ$.

Next, the characteristics seen in the idealized simulations can be used to interpret the observed

TABLE 1. Summary of the virtual temperature profiles for the idealized simulations, representing typical, stable, and almost neutral atmospheric conditions.

Height (m)	Profile		
	Typical (K)	Stable (K)	Almost neutral (K)
0	292.9	278.2	293.1
1500	297.2	291.2	294.2
3000	303.2	303.2	295.2
5500	313.2	313.2	297.2

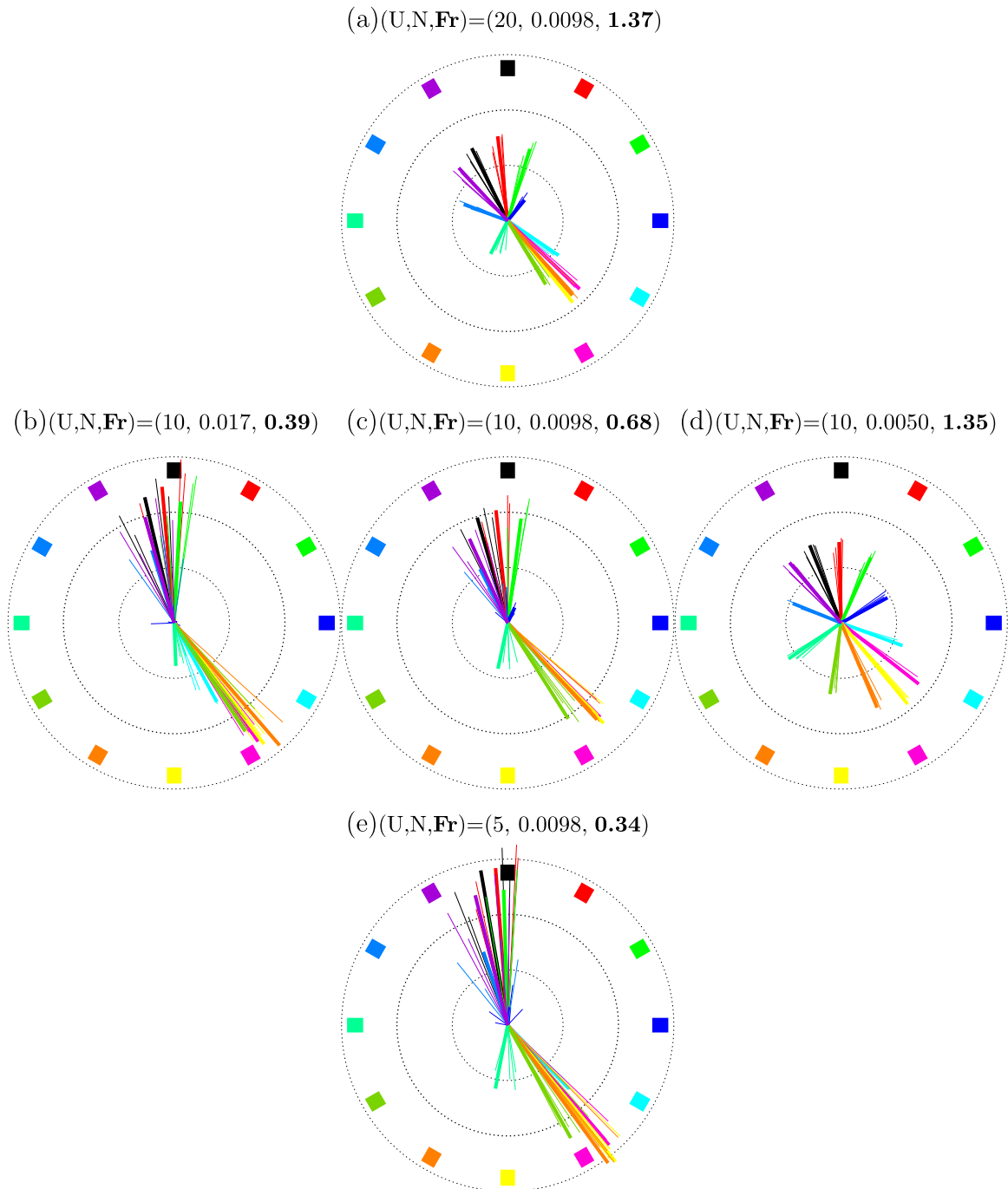


FIG. 8. Results from the idealized simulations in the form of polar plots showing wind velocities around the site (see text for details): (a) for 20 m s^{-1} forcing for typical stability; for 10 m s^{-1} forcing for (b)–(d) stable conditions, typical stability, and almost neutral conditions, respectively; and (e) for 5 m s^{-1} forcing for typical stability. The geostrophic wind speed, Brunt–Väisälä frequency, and Fr are indicated above each plot. The geostrophic wind speed increases moving upward from (e) to (a). The Brunt–Väisälä frequency decreases moving from left to right from (b) to (d). The Froude number increases both when moving upward and when moving from left to the right.

characteristics at the wind farm site. In Figs. 2 and 3 the Froude number tends to increase with an increase in the large-scale geostrophic wind speed, as expected considering the role of U in (1). Looking over all wind

directions, for a given wind speed there is a variation in the Froude number due to the variation of the static stability N^2 . In contrast, the Froude number has a much less predictable relationship with the wind farm site

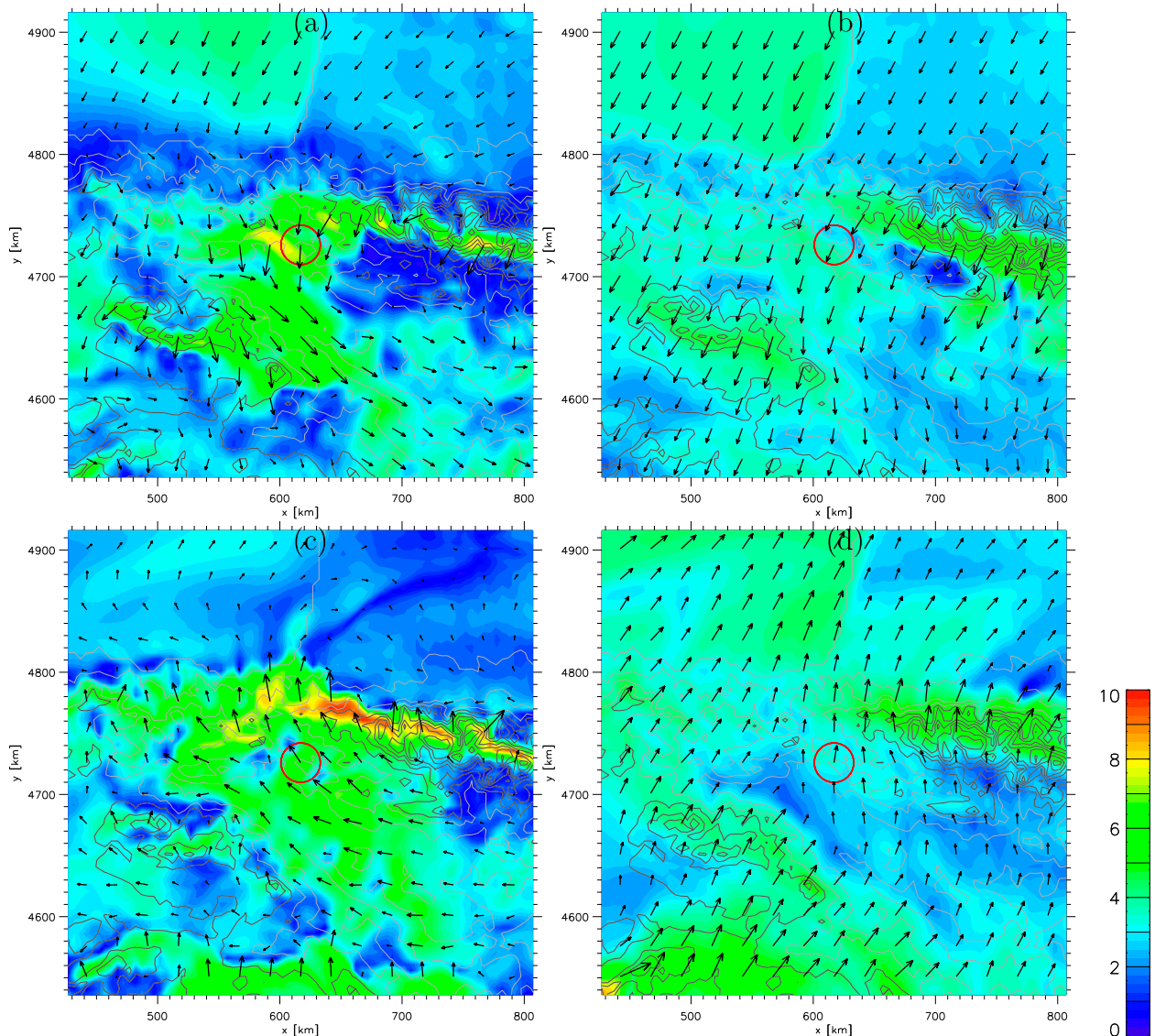


FIG. 9. Maps showing the winds at 50 m AGL for (left) stable and (right) near-neutral conditions for geostrophic wind forcing of 10 m s^{-1} from (a),(b) 60° and from (c),(d) 240° . The center of the red circle indicates the location of the wind farm site. The color scale to the right indicates the wind speed in meters per second.

surface winds (Fig. 7). This is because, as seen from the idealized modeling, the low Froude number flow is associated with a large enhancement of surface wind speed. This leads to the nonintuitive situation of weaker geostrophic winds causing stronger surface winds at the farm site.

The simulation using wind classes in Fig. 6 shows a consistent picture of scrambled Froude number values in the north-northwest and south-southeastern sectors, which are also the most frequent wind directions. In Fig. 7, for westerly winds at the wind farm site, the Froude number is predominantly relatively high. This can be understood by noting in Fig. 8a that westerly wind at the

wind farm site is only achieved by high Froude number flow with geostrophic wind direction in the range $270^\circ \leq \theta \leq 300^\circ$. Similarly, easterly winds at the wind farm site are only achieved at the site for high Froude number flow with geostrophic wind direction in the range $90^\circ \leq \theta \leq 120^\circ$.

Referring to Fig. 2, the mean wind speed in the range of geostrophic wind direction $270^\circ \leq \theta \leq 300^\circ$ is 9.3 m s^{-1} , the mean Froude number is 0.61, and the frequency of occurrence is 10.2%. For the range of geostrophic wind direction $90^\circ \leq \theta \leq 120^\circ$, the mean wind speed is 5.9 m s^{-1} , the mean Froude number is 0.49, and the frequency of occurrence is 6.6%. The occurrence of westerly wind (albeit low wind speed) at the

site can be explained by the occurrence of the suitable large-scale flow situation, whereas the very rare occurrence of easterly wind (again low wind speed) is explained by the weaker (lower Froude number) and less frequent occurrence of the required large-scale flow situation. The plots in Figs. 2, 3, 6, and 7 are presented with Cartesian axes, instead of the polar axes, because the low wind speed data points are more separated and more clearly seen.

This analysis demonstrates that the wind class method allows a very clean and straightforward definition of atmospheric flow over the entire modeling domain, which would not be possible using weather typing. Weather types typically represent situations with curvature to the flow, due to different and realistic placements of high and low pressure centers, giving an additional and unnecessary degree of complexity to the downscaling.

4. Postprocessing mesoscale model results for wind resource assessment

a. Motivation

Mesoscale modeling is able to determine the variation of wind fields at scales down to 2–10 times the model grid spacing (e.g., scales of 10–50 km for model grid spacing of 5 km) (Skamarock 2004). Therefore, it must be reconciled that there will be disagreement between the modeled and measured wind statistics, because of the missing variance at smaller scales, including the subgrid scale. In nature, the boundary layer winds are affected by the local topography, by (i) orography, giving rise to speedup effects on hilltops and (ii) surface roughness and surface roughness changes, giving rise to variations in the vertical profile of the wind speed and the internal boundary layers. In addition, the boundary layer winds are affected by the stability of the atmosphere. Similarly, in the mesoscale model, the winds are affected by the topography and stability as it is represented in the model.

In this study, to compare modeled winds to measured winds, a process known as wind climate generalization is used. It entails removing the topography effects influencing the modeled wind. This is done using the methods described below. For the measured winds the methods are similar and are described in some detail in Troen and Petersen (1989).

There are two important assumptions to state explicitly before describing the details of the method:

- 1) Topography effects influencing the winds are considered in terms of independent orographic effects and surface roughness effects. The total effect is a linear combination of the individual effects.

- 2) A neutral boundary layer is assumed for the determination of the orographic effects and for the calculation of surface winds for new surface roughness lengths.

b. Generalization of gridpoint wind data

In this section a single method for calculating the generalized wind climate statistics from mesoscale modeling is presented. The starting point for the calculation of the generalized wind climate profiles is the mesoscale model wind speed and direction, $u_s(z_i)$ and α_s , for predetermined heights z_i .

Typically, these heights are 10, 25, 50, 100, and 200 m. The wind speeds are determined by linear interpolation of wind speed at model levels from mesoscale simulations for each wind class simulation. If the level is below the lowest model level, then extrapolation is performed using a Monin–Obukhov similarity function.

The first step in the generalization process is to account for how the mesoscale model description of topography impacts the local flow. This is done by modeling the local flow perturbations, which are functions of wind direction and height above the surface. The linear flow model LINCOM (Astrup et al. 1996) is the basis of the calculation of the local flow perturbation on wind speed due to orography δA_{oro} , defined by

$$\delta A_{\text{oro}} = \frac{|u_s|}{|u_{\text{flat}}|} - 1, \quad (3)$$

where $|u_s|$ is wind speed in the mesoscale model at a single height and $|u_{\text{flat}}|$ is the calculated wind speed at the same height for flat terrain. Positive δA_{oro} corresponds to an orographic speedup. The orography changes the local wind direction also, so there is a local flow perturbation on wind direction, namely $\delta \alpha_{\text{oro}}$:

$$\delta \alpha_{\text{oro}} = \alpha_s - \alpha_{\text{flat}}, \quad (4)$$

where α_{flat} is the wind direction for flat terrain and α_s is the wind direction in the mesoscale model. Both δA_{oro} and $\delta \alpha_{\text{oro}}$ are dependent on height. The calculation of the local flow perturbations uses the orography grid used in the mesoscale simulations. The orographic flow perturbation to the wind speed and directional, δA_{oro} and $\delta \alpha_{\text{oro}}$, are calculated assuming linear, shear flow, and neutral stratification.

Similarly, there is a local flow perturbation on wind speed due to roughness length variations. The local flow perturbation due to roughness δA_{rou} relates the wind speed in the mesoscale model to the wind speed for a uniform roughness set at the height-independent upstream roughness length $\hat{z}_0$ according to

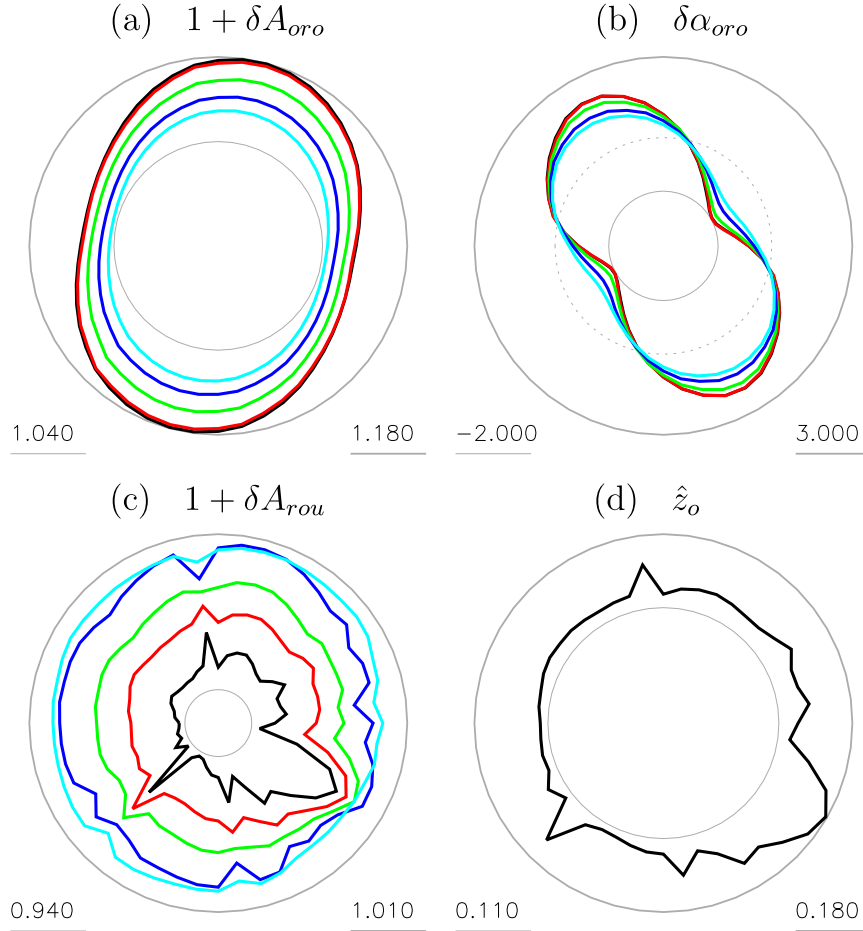


FIG. 10. Mesoscale generalization parameters for Alaiz: (a) orography factor, (b) orography turning, (c) roughness factor, and (d) effective upstream roughness (m). The minimum (maximum) values for the radial scale are shown by the thin (thick) gray lines. Values for five heights AGL for (a)–(c) are 10 m (black), 25 m (red), 50 m (green), 100 m (dark blue), and 200 m (light blue).

$$\delta A_{\text{rou}} = \frac{|u_s|}{|u_{\hat{z}_0}|} - 1, \quad (5)$$

$$|\hat{u}| = \frac{|u_s|}{(1 + \delta A_{\text{oro}})(1 + \delta A_{\text{rou}})} \quad (6)$$

where $|u_s|$ is the wind speed in the mesoscale model and $|u_{\hat{z}_0}|$ is the calculated wind speed expected given a uniform upstream roughness $\hat{z}_0$. The height-independent upstream roughness length, $\hat{z}_0$, is a function of the wind direction.

The wind profile at a specific location is not determined by the roughness at the site alone but is influenced by the roughness upstream of the site. If there are changes in the upstream roughness length, then the wind profile will be modified by the development of internal boundary layers (Sempreviva et al. 1990). The intermediate wind speed, accounting for mesoscale orography and roughness changes, is

and the intermediate direction is given by

$$\hat{\alpha} = \alpha_s - \delta\alpha. \quad (7)$$

Instead of local flow perturbations on wind speed, it is convenient to use the generalization factors. These are defined as $F_{\text{oro}} = 1 + \delta A_{\text{oro}}$ and $F_{\text{rou}} = 1 + \delta A_{\text{rou}}$ for the orographic and roughness generalization factors, respectively.

Figure 10 shows a polar representation of the generalization parameters for the wind farm site. The orography speedup F_{oro} is shown in Fig. 10a. The strongest speedup is seen for the N and NNE and S and SSW wind directions. For these direction sectors a speedup of 18%

is exhibited at 10 m above surface level. Note that the speedup decreases with height. The orographic local flow direction perturbations are shown in Fig. 10b. Minimum turning (no turning) occurs for the maximum and minimum speedup directions, where the wind direction is aligned with the geometrical axes of the orography. Maximum turning occurs for the wind directions bisecting these axes.

The roughness generalization factor, $1 + \delta A_{\text{rou}}$, and effective upstream roughness $\hat{z}_0$ are shown in Figs. 10c and 10d, respectively. The roughness generalization factor is also a function of height above the surface. Where there is relatively homogeneous upstream roughness length in the mesoscale model, the roughness generalization factors at different heights tend to be similar and close to 1. For more heterogeneous upstream roughness, roughness generalization factors may vary significantly with height. This is due to internal boundary layers and the wind profile departing from logarithmic. For example, for a roughness change from low to high as the location of interest is approached, at low heights the winds will tend toward equilibrium for the high roughness more rapidly; therefore, lower roughness generalization factors are expected for lower heights. Conversely, for a roughness change from high to low as the location of interest is approached, higher roughness generalization factors for lower heights are expected. In the situation in which there are multiple changes in upstream roughness, the change in the generalization factor with height is complex and reflects the different internal boundary layers present at different heights.

It is worth noting that the method described in Yu et al. (2006) would not allow this kind of consideration of roughness change. Figures 10c and 10d show that these factors can be significant.

c. Calculating wind climate data

At this stage of the process, we have for each single grid point for each simulation the wind speed $\hat{u}_i$ and direction $\hat{\alpha}_i$ for the heights z_i for the gridpoint-dependent single roughness length $\hat{z}_0$. The final stage in the generalization of the mesoscale winds is to determine the wind speed and direction for specific roughness length z_{0j} . This is done in three steps. First, using the geostrophic drag law for neutral conditions the nominal geostrophic wind speeds $\hat{G}_i$ and wind directions $\hat{\alpha}_{G_i}$, are calculated, using the intermediate wind speed $\hat{u}_i$ and wind direction $\hat{\alpha}_i$ at height z_i , according to

$$\hat{G}_i = \frac{\hat{u}_{*i}}{\kappa} \sqrt{\left(\ln \frac{\hat{u}_{*i}}{|f|\hat{z}_0} - A \right)^2 + B^2} \quad (8)$$

and

$$\hat{\alpha}_{G_i} = \hat{\alpha}_i + \arcsin \left(\frac{B u_{*i}}{\kappa \hat{G}_i} \right), \quad (9)$$

with the intermediate friction velocity $\hat{u}_{*i}$ given by

$$\hat{u}_{*i} = \frac{\hat{u}_i(z_i) \kappa}{\ln(z_i/\hat{z}_0)}, \quad (10)$$

where κ is the von Kármán constant and for neutral conditions A and B are set to 1.8 and 4.5 (Northern Hemisphere) or -4.5 (Southern Hemisphere), respectively. Note that $\hat{\alpha}_{G_i} - \hat{\alpha}_i$ is the angle between the geostrophic wind direction and the intermediate near-surface wind direction, and that this is positive in the Northern Hemisphere (i.e., the geostrophic wind direction is turned clockwise relative to the surface wind) and the reverse is true in the Southern Hemisphere. Second, the generalized friction velocity, $u_{*g}(z_{0j})$, is calculated using the geostrophic drag law iteratively, where z_{0j} is the roughness length value. Third, using the logarithmic profile, the generalized wind speed, $u_g(z_i, z_{0j})$, is calculated using

$$u_g(z_i, z_{0j}) = \frac{u_{*g}(z_{0j})}{\kappa} \ln \frac{z_i}{z_{0j}} \quad (11)$$

and the generalized wind direction, $\alpha_g(z_i, z_{0j})$, is calculated using

$$\alpha_g(z_i, z_{0j}) = \hat{\alpha}_{G_i} - \arcsin \frac{B u_{*g}}{\kappa \hat{G}_i}. \quad (12)$$

This is done for each of the wind class simulations. At this point it would be convenient to carry out direction-sector-dependent fitting of the generalized wind to the Weibull distribution, using the wind class generalized winds and their associated wind class frequencies. The Weibull distribution, which is commonly used in wind climate statistics, is described by two parameters, a scale parameter A_W and a shape parameter k_W . However, the number of simulations is relatively small in comparison with measurement data, for which Weibull fitting is normally performed, so the fitting is poor. The problem is addressed by a method that interpolates the generalized winds across similar wind classes. Here, interpolation is used to yield a wind speed and direction dataset 5 times larger than the simulation dataset. With the enlarged dataset, the direction-sector-dependent Weibull parameters A_W and k_W can be determined for any or all grid points in the modeling domain.

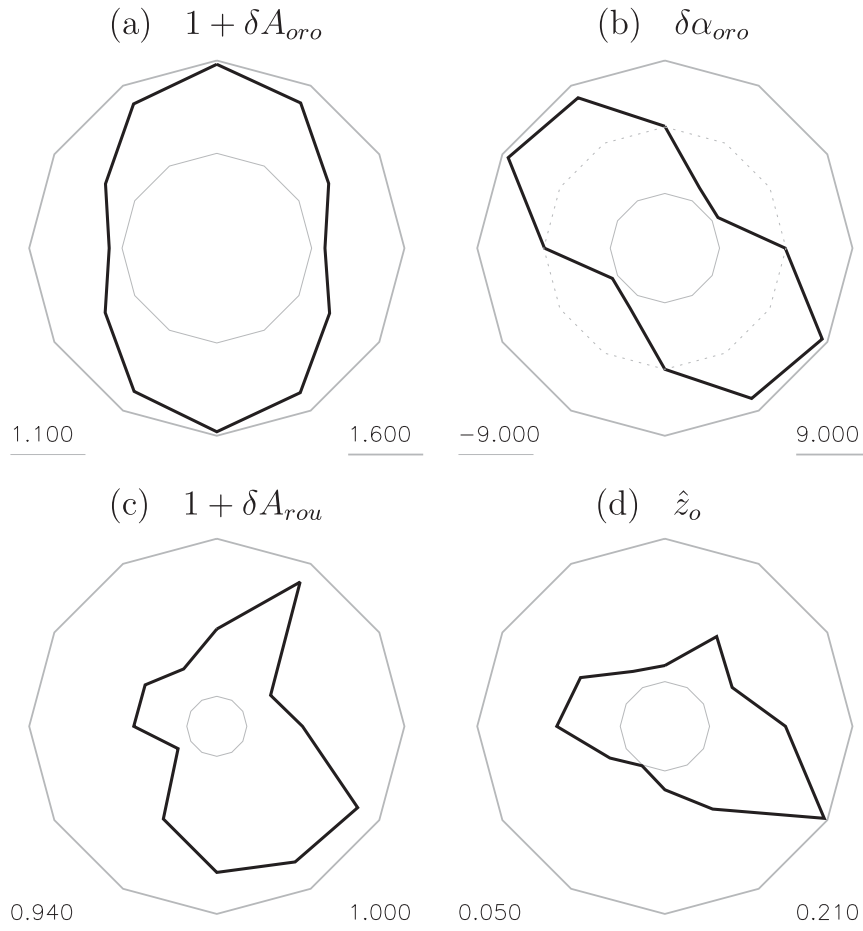


FIG. 11. As in Fig. 10, but for the microscale generalization parameters at 55 m AGL.

The end result of the generalized wind climate calculation is generalized wind climate data. In practice, this is a so-called WASP lib-file (Troen and Petersen 1989). The WASP lib-file contains tables of direction-sector-dependent A_W and k_W Weibull parameters for a set of heights above the surface and a set of surface roughness lengths. The heights typically are 10, 25, 50, 100, and 200 m [i.e., values z_i in Eqs. (11) and (12)] and the roughness lengths are typically 0.0002, 0.03, 0.1, 0.4, and 1.5 m [i.e., values z_{0j} in Eqs. (11) and (12)]. The direction sector frequencies are also given, typically using 12 sectors of equal size.

5. Application and verification

In this section the generalized wind climate will be applied to the wind farm site using high-resolution topographical data of the surroundings. The purpose of this simulation is to demonstrate the application of the mesoscale-model-derived data in a microscale model. The downscaling results using the mesoscale to

microscale model chain will be compared to measurements at the wind farm site.

Models within WASP (Troen and Petersen 1989) introduce the effects of microscale terrain into the generalized wind climate. The process is the generalization in reverse. Figure 11 is analogous to Fig. 10 but this time for the orography and roughness description at very high resolution and at the height of the anemometer (55 m AGL). The topography information, which forms the basis of the generalization of the observed winds, is in vector format. Therefore, a resolution in terms of a horizontal grid spacing cannot be stated for the topographic data. However, an accuracy for the placement of topographic features on the order of 10s of meters is estimated. When the high-resolution orography is taken into account, a speedup of 60% is estimated. The models LINCOM and WASP are very similar, with only some differences in the implementation of the roughness correction and the geometry of the model grid. WASP is very widely used in the wind energy community. It is used to generalize wind measurements and to predict

TABLE 2. Summary of error (%) for mean wind speed and power density estimates using the different postprocessing methods from mesoscale model output to site conditions.

Approach		Wind speed error (%)	Wind power density error (%)
A1	No generalization and no microscale effects	−25.4	−57.1
A2	No generalization and no microscale effects, but site roughness	−28.6	−62.1
A3	No generalization and no microscale effects	21.3	70.6
A4	Generalization and no microscale effects	−33.5	−70.7
A5	Generalization and no microscale effects, but site roughness	−36.1	−74.2
A6	Generalization and microscale effects	5.00	9.62

wind climate at wind turbine sites. If the type of turbine is known, WASP can also calculate the expected mean annual energy production.

To demonstrate the skill of the postprocessing generalization, other alternative methods of applying mesoscale model output to estimate conditions have also been tested. Direct application (approach A1) of mesoscale output to site conditions entails interpolation of the model-level winds to the height of the anemometer (55 m AGL), at the model grid point over the site. In most cases the mesoscale roughness length for the grid point at the site is not the same as the roughness length at the site. Therefore, a very straightforward correction of the mesoscale winds is to account for this difference in roughness length (approach A2). This can be done using the geostrophic drag law. It should be remembered that this roughness correction only accounts for roughness length immediately at the site, and that any roughness changes in the vicinity, leading to internal boundary layers, will not be accounted for. The next level of correction is to account for the local microscale effects at the site (approach A3), that is, the orographic speedup and effects of roughness and roughness change. Another approach to correction entails applying generalization corrections for the mesoscale description of the terrain to the mesoscale model output (approach A4), but not making any adjustments for the local microscale effects. Next, microscale effects limited to the impact of the immediate roughness at the site are accounted for (approach A5). Finally, the method through which generalization corrections and microscale terrain corrections are applied is used (approach A6).

The results of the six approaches are given in Table 2. The results are given in terms of the mean wind speed and mean nominal power density at 55 m AGL at the site. The mean nominal wind power density is given by

$$\bar{P} = \frac{1}{2} \rho_s \bar{u}^3, \quad (13)$$

where ρ_s is a constant standard air density, set to 1.225 kg m^{-3} . Use of the constant standard density

means that the actual power density at the site will differ from the nominal power density. However, in this investigation the same standard density is used for the measurements and modeling results. So the comparison is consistent and fair. Comparison of mean nominal wind power density is a good indicator of how well the distribution of the wind speed at the site is captured. Variation across the six methods is seen more strongly when looking at the nominal wind power density. It should be noted that for an actual resource assessment at the site, an annual mean air density adjusted according to the site mean temperature and pressure would be required.

The direct application of mesoscale model output to site conditions (approach A1) yields poor results. The mean wind speed and mean power density are underestimated by over 20% and 50%, respectively. Applying an immediate roughness length correction for the site (approach A2) does not improve the results. This actually lowers the already underestimated value by a few percent, since the roughness length for mesoscale model grid points is lower than the actual roughness length at the site.

Application of microscale corrections only (approach A3) gives overestimations of the mean wind speed and power density of approximately 25% and 70%, respectively. This is mainly due to the orographic speedup effect being added by the microscale corrections. However, the orographic speedup effect is partly already present in the mesoscale model output. In this case the orographic speedup effect is accounted for twice.

When the mesoscale terrain effects are corrected for through generalization (approach A4), the mean wind speed and power density estimations are severely underestimated (approximately 30% and 70%, respectively). The underestimate is larger than in the direct application case, because the mesoscale terrain orographic speedup effect has now been removed. Accounting for immediate roughness (approach A5), as before, decreases the underestimated value even further (by a few percent). Finally, carrying out mesoscale terrain corrections, through generalization, and applying microscale terrain corrections (approach A6) gives the

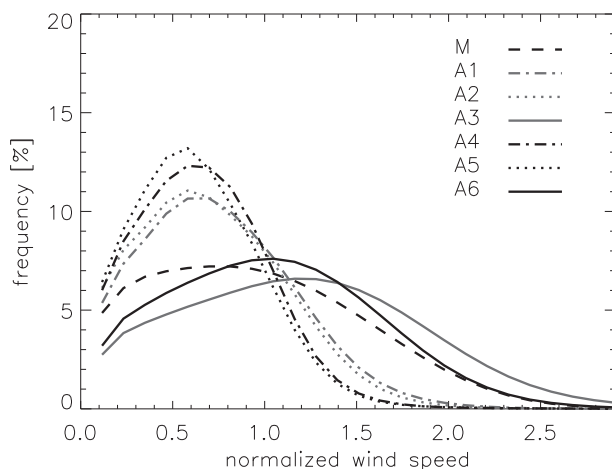


FIG. 12. Wind speed frequency distributions for the Alaiz site as given by measurements (M) and postprocessing methods (A1–A6; Table 2), as indicated by the legend.

best estimates, with the wind mean speed and power density only modestly overestimated by approximately 5% and 10%, respectively.

Figure 12 shows the omnidirectional wind speed distributions derived from measurements and postprocessing approaches A1–A6. The distributions are derived from the frequency-weighted direction-sector-dependent Weibull distributions. The difference in the distributions of approaches A1 and A2 show the impacts of accounting for the site roughness, which is a slight downward scaling of wind speeds. This is consistent with the site surface roughness being higher than the surface roughness at the site location in the mesoscale model. The same feature is also true when comparing the wind speed distribution for approaches A4 and A5. The difference in the distributions of approaches A1 and A3 shows the impact of accounting for the microscale roughness and orography. This is a large upward scaling of the wind speed, with the dominant effect being the orographic speedup. Approach A3 yields a wind speed distribution with too high frequencies at high wind speeds. This is because the orographic speedup is accounted for twice. In approach A6, the roughness and orographic (speedup) effects in the mesoscale modeling are removed (shown in the difference in the distributions for approaches A1 and A4) and the microscale roughness and orographic (speedup) effects added. The distribution for approach A6 is closest to the distribution derived from the measurements. The model wind speed is somewhat higher for approach A6 compared to the measurements but the distributions begin to approach each other in the upper tail of the distribution. This is an important part of the distribution for wind energy applications.

6. Discussion

a. Time period

In this study only data for 2001 have been used. This was due to the availability of measurement data at the site. However, there is no difficulty in extending the methodology to cover longer periods. Typically, for wind resource assessments, a 30-yr wind climatology is calculated. To do this, the wind classes are defined following the method outlined in section 2 using 30 yr of reanalysis data. This ensures that the wind classes are appropriate for capturing the variation in wind conditions occurring over the entire period, because the wind classes will be spread over the climatological distribution of wind speeds for each sector.

Once the climatological wind classes are determined, the methodology is able to determine site wind statistics for a shorter, specified measurement period. This is very valuable for validation studies, where it is important that the modeling result covers the same period as the measurements. The procedure involves recalculating the wind class frequencies using the reanalysis data for the measurement period only and then repeating the postprocessing with the new wind class frequencies. This method allows for the recalculation of wind statistics during numerous time periods without the need for repeating mesoscale simulations. Recalculating the wind class frequencies for a shorter period will result in a different and unequal set of frequencies within direction sectors. The purpose of the equal frequency wind classes was to distribute the computer resources efficiently over the distribution of wind conditions, but since no repeated mesoscale modeling is required, this consideration is no longer a requirement. However, it may impact the accuracy of the methodology slightly. This impact has not been studied as yet but could be an interesting area for further investigation.

b. Sources of error and uncertainty measures

There are many sources of uncertainty within the methodology. It is very difficult to isolate the uncertainty sources because verification is only possible against wind measurements and not at any midway stage in the methodology. The first source of uncertainty is within the description of the large-scale meteorological conditions. The NCEP–NCAR reanalysis accuracy is better in regions of high-density observations. In regions with scarcer measurements the accuracy is degraded. Errors will be introduced by the representation of the large-scale meteorological conditions by a finite number of wind classes. In section 2, the sensitivity of accurately obtaining wind speed distributions to wind class number was examined.

The level of uncertainty introduced by the wind classes is commensurate with other uncertainties in the model chain. Therefore, increasing the number of wind classes does not necessarily decrease the error of the final wind climate prediction. Errors in the wind velocities may lead directly to errors in wind resources. More indirectly, wind velocity errors can also lead to a different interaction of the flow with terrain, as the Froude number will also be modified. Similarly, errors in temperature profiles will lead to errors in stability and Froude number, and may also cause erroneous flow behavior in complex terrain. Capturing the Froude number correctly has been shown to be very important for the Alaiz site from the results shown in [section 3c](#). Errors are also associated with the topographical description. The surface elevation errors may be due to insufficient spatial resolution. One of the most serious consequences of an error in the orography is the underrepresentation of high terrain, which may lead to incorrect interactions by the flow with terrain. For example, for a low Froude number situation, the flow in the model may not be blocked as expected because the terrain creating a barrier is lower in the model than it should be. For the Alaiz cases, underrepresentation of high terrain would lead to a reduction in the gap flow behavior and the modeling would miss the concentration of wind directions and the enhancement of wind speeds. Errors in surface roughness may be due to the insufficient spatial resolution of land-use data and through the incorrect estimation of the roughness length. Errors will also be introduced through the mesoscale modeling itself. The KAMM simulations used in this methodology assume uniform and steady atmospheric forcing for each wind class, thus any wind features due to transient and spatially varying forcings are not accounted for.

The assumption of a neutral boundary layer to determine topographic effects in the postprocessing may be a less serious source of error compared to some of the sources outlined above, because the objective of the method is to determine climatological conditions where the mean situation is close to neutral. The assumption would have a more serious impact on error if the postprocessing was used on individual days, with stability deviating from neutral. The assumption concerning the independent treatment of orographic and surface roughness effects has an impact that at this time is very difficult to determine. The assessment would require a comparison of terrain effects with a model using the dependent treatment of orographic and surface roughness effects. Candidate models for this would include Reynolds-averaged Navier–Stokes equations based computational fluid dynamics models (RANS models).

The error for the Alaiz site, when using the generalization procedure, is in line with the verification studies

performed routinely as part of numerical wind atlas studies. Two measures of uncertainty are used. Starting with [Frank et al. \(2001\)](#), the uncertainty was determined for three numerical wind atlas studies using the mesoscale model KAMM in Denmark, Ireland, and an area covering both northern Portugal and northwestern Spain. There the uncertainty was determined by the RMS value of the relative differences in the modeled and measured annual mean wind speeds and power densities for a number of sites:

$$\text{RMS}_{\text{diff}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left[\frac{2(P_i - O_i)}{P_i + O_i} \right]^2}, \quad (14)$$

where N is the number of verification stations and P_i and O_i are the predicted and observed annual mean quantities, respectively, for the i th station. At 50 m AGL, the RMS_{diff} error on wind speed was 5% for Denmark (21 sites), 7% for Ireland (29 sites), and 10% for northern Portugal and northwestern Spain (13 sites). The increased complexity of terrain in the region and the increased importance of thermal circulations over northern Portugal and northwestern Spain can explain the larger error.

The error has more recently been expressed in the form of the mean absolute error MAE:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{P_i - O_i}{O_i} \right|. \quad (15)$$

For the Egyptian wind atlas ([Mortensen et al. 2005](#)), the mean absolute error on wind speed was determined to be 7% for generalized wind climates at 18 sites. For the Indian wind atlas ([Sreevalsan et al. 2010](#)), the mean absolute error was 13%, based on generalized wind climates at 42 sites. More recently for the wind atlas for the region of Dongbei ([Badger et al. 2011](#)), which comprises China's three northeastern provinces, the mean absolute error was determined to be 8%, based on generalized wind climates at nine sites. For a wind resource studies for South Africa and Mali using the same methodology, the mean absolute errors have been determined to be 6% and 4%, respectively.

c. Other applications

The preprocessing and postprocessing methodologies described in this paper also have applications outside of mesoscale modeling for wind resources. For example, the wind class method has been employed in wind climate assessments using synthetic aperture radar (SAR) satellite images ([Badger et al. 2010](#)). The SAR images can be processed to give an estimate of a near-instantaneous

two-dimensional wind field at 10 m above sea level. Wind classes have been used to determine which satellite images should be selected for processing, so a representative range of atmospheric conditions is captured. Available SAR images can be categorized according to the wind class occurring at the time the image was obtained. If a selection strategy is not used, there is a risk that scenes will not be climatologically representative and a bias may be introduced into the wind climate estimates. In addition, the wind class frequency information is used to determine the appropriate weighting of SAR-image-derived winds for the calculation of the wind climate.

Another application of the generalization methodology is in the estimation and verification of extreme wind climate, in terms of the 50-yr return wind. In [Larsén et al. \(2012\)](#), storm episodes were simulated using the Weather Research and Forecasting (WRF) mesoscale model. To compare the extreme wind statistics obtained from the simulations with extreme wind statistics obtained using measurements, a generalization step is necessary to take account of the orographic speedup and roughness length effects in the simulated winds and the measurement.

d. Alternative methods

It is possible to link mesoscale model output to microscale models without generalization. In [Yu et al. \(2006\)](#), the link was made by using a perturbation elevation map for the microscale modeling (the perturbation elevation map is the elevation minus the mesoscale model field for elevation). This indeed makes sense within a linear framework (i.e., to subtract orography that is already present in the mesoscale model). Unlike the method outlined in this paper, which is also based on a linear framework, the [Yu et al. \(2006\)](#) method has the disadvantage that the mesoscale wind data need to be accompanied by the mesoscale orography, if any further downscaling is to be performed using another microscale elevation dataset, for example at higher resolution. Also, it is uncertain how this approach can best account for the roughness changes present in the mesoscale model description of terrain.

The wind class method has been developed within a statistical–dynamical downscaling methodology with mesoscale modeling employing steady and uniform (geostrophic) forcing. However, another potential application of the wind class methodology is within a dynamical downscaling approach. Noncontiguous periods can be simulated with mesoscale models to calculate wind climatologies, as in [Tammelin et al. \(2013\)](#), in which 48 month-long simulations were made; for each calendar month, that month from four different years was selected. In [Tammelin et al. \(2013\)](#), the choice of

each month's four years is based on the best representativeness of the combinations of 4 months compared to the period 1989–2007, using data from the European Centre for Medium-Range Weather Forecasts (ECMWF) interim reanalysis (ERA-Interim).

It would be of interest to compare methods for selecting climate representative periods based on wind class with alternative methods such as those in [Tammelin et al. \(2013\)](#) and [Rife et al. \(2013\)](#). In [Rife et al. \(2013\)](#), a Monte Carlo method is applied to efficiently select a subset of days that maintain the same climate statistics, in terms of probably density function, as a long climate period.

Finally, it is of interest to interpret the work presented in [Jiménez et al. \(2008\)](#) in terms of the generalization framework described here. In [Jiménez et al. \(2008\)](#) an evaluation of mesoscale simulations at regional scale, rather than at specific sites, is developed. The standpoint is that the regionalization of model output allows for fairer comparisons of models and measurements, by an aggregation over region type, where the region is defined by wind characteristics. The advantage comes from a filtering out of “noise associated with local effects.” However, it could alternatively be argued that it is exactly these local effects that are of interest, especially from the viewpoint of wind energy and siting of wind turbines, ventures that seek to exploit local effects.

7. Conclusions

The mesoscale flow characteristics at a wind farm site have been examined using the wind class method and idealized mesoscale modeling. It has been shown that much of the wind characteristics at the site can be captured by mesoscale modeling, in particular the strong concentration of winds into the northerly and southerly sectors. The idealized model runs show that the Froude number plays an important role in determining the wind conditions at the site. It confirms the importance of a large-scale Froude number as an important aspect of a wind classification system, which can be determined from reanalysis data.

A method for the application of mesoscale modeling output to site conditions has been described and tested for the site. The basis for the method involves the post-processing of mesoscale model output to create generalized wind climate statistics. The mean wind speed, mean power density, and the omnidirectional wind speed distribution at 55 m AGL at the site were calculated using the generalized wind climate and the WAsP model and compared to measurements. Different, simpler, routes from mesoscale modeling output to site conditions were also tested and found to result in much poorer

performance when compared with the generalization postprocessing and WAsP application method.

The generalization postprocessing method has been used extensively in a number of wind resource assessment studies, so-called numerical wind atlas studies, with success. The postprocessing allows a proper verification to be carried out, in which wind climate estimates derived from mesoscale modeling and measurements can be compared. Without the postprocessing step, no verification is possible, because the surface description within the model does not agree with reality, and therefore model winds will not agree with measured winds, except perhaps in extremely simple terrain or over water far from coasts.

When high-quality wind measurement data are available, the agreement between modeled and measured wind climates is between 5% and 15% depending on terrain complexity, using the mean absolute error for the estimated wind speed at 50 m AGL.

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